

BIL 481 Software Engineering

Project Assignment 1

Project Concept Development with Gemini

Released Monday, 5 October 2026

Due Sunday, 11 October 2026, 23:59
(Türkiye time, UTC+3)

Instructor Dr. Çiğdem Avcı

TA Ömer Kılınç

Overview. This is the first assignment of your course project. Working with your existing project group, you will create **Concept Notes** outlining your project's problem, intended users, requirements, and preliminary design and implementation plan. **Gemini is central to this assignment:** use it to draft, critique, compare, and refine your proposal. Evaluate its suggestions together and make the final project decisions as a group.

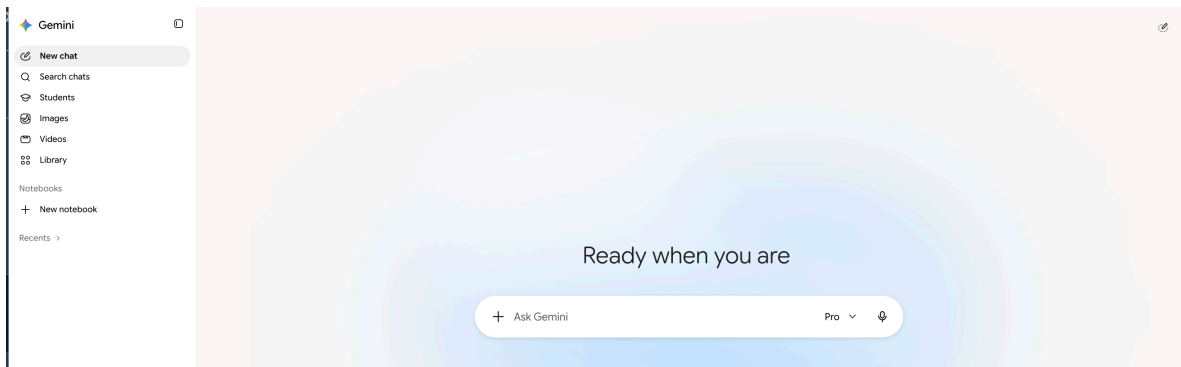
1 Setup

Use **Google AI Plus**, or your existing **Google AI Pro or Ultra** subscription, for **Assignment 1**. Work in the standard Gemini web app; this assignment does not use Spark. Work with your **existing project group**. Each group must use **one** Gemini notebook and chat, one Google Drive project folder, one Concept Notes document, and one public GitHub fork. Designate one member to operate the Gemini chat while all members participate in discussions and decisions.

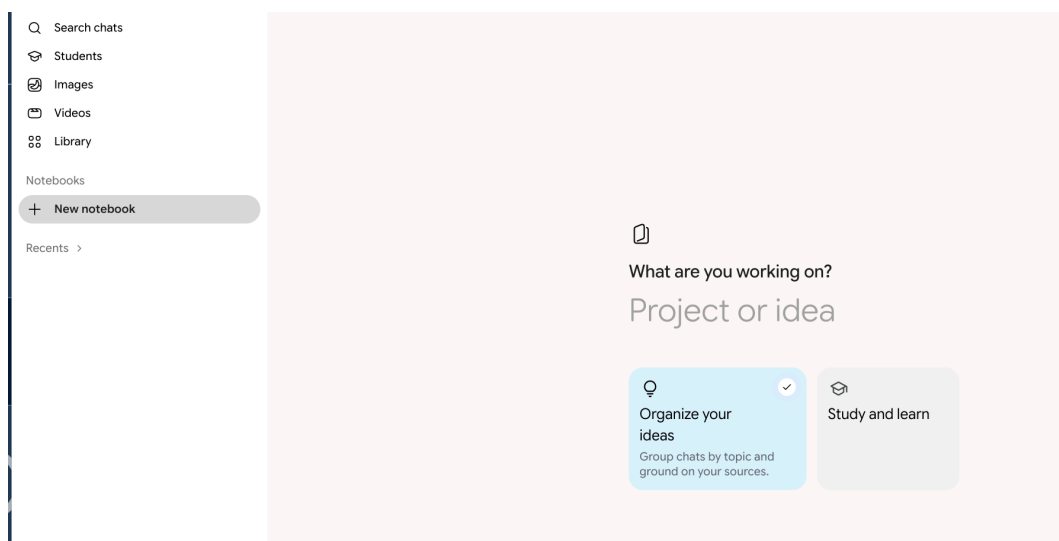
1. All group members must use their own eligible personal Google account. If you already have **Google AI Pro or Ultra**, keep your existing subscription; **you do not need to downgrade**. Otherwise, activate **Google AI Plus** through the [Google student offer](#). Eligible university students aged 18 or over in Türkiye receive the first 12 months free. Complete the student verification shown during signup.
2. The designated member signs in to [Gemini](#) and Google Drive using the **same personal Google account** with Plus, Pro, or Ultra active. Keep **Keep Activity** enabled so the notebook chat is saved. This member creates and maintains the group's notebook, chat, folder, and document.
3. Under **Notebooks** in the left sidebar, select **New notebook**. Choose **Organize your ideas**, enter BIL481-Project as the name, and select the arrow to create it.
4. Open BIL481-Project and start **one chat** for your group's Assignment 1 work. Use this same chat for setup, document creation, the initial project prompt, critique, comparison, and every document update.

2 Setup walkthrough: create the notebook

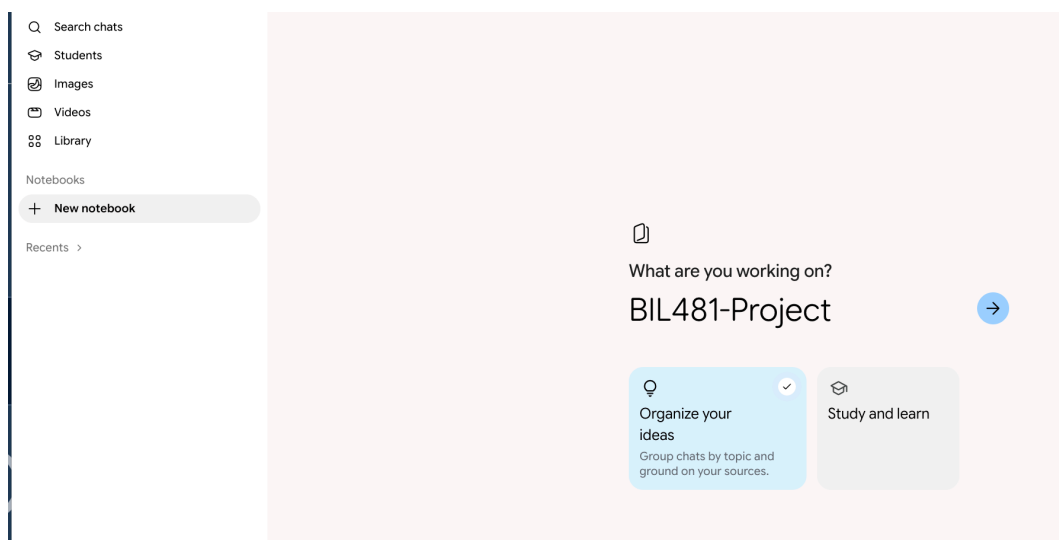
Follow Setup steps 3–4 using the designated member's account.



Screenshot 1. In Gemini's left sidebar, under **Notebooks**, select **New notebook**.



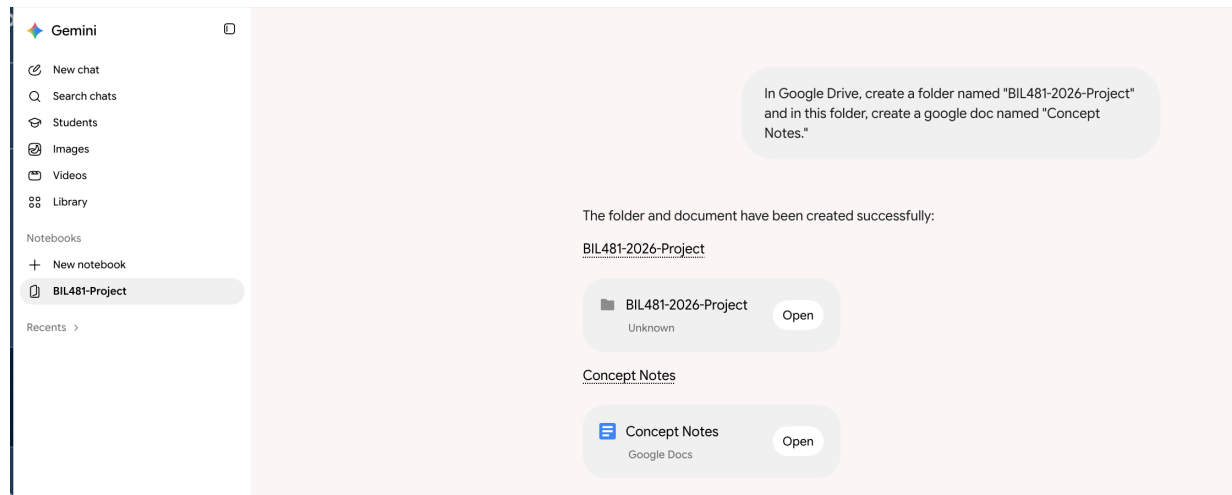
Screenshot 2. Choose **Organize your ideas** for the project notebook.



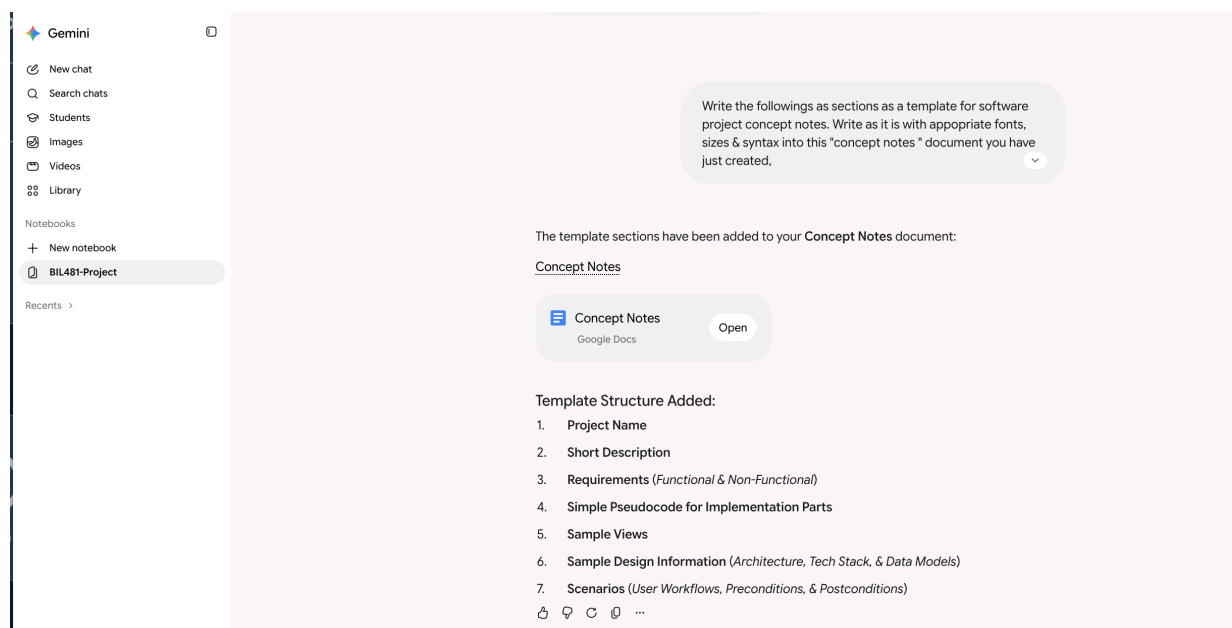
Screenshot 3. Enter BIL481-Project, then select the arrow to create the notebook.

3 Setup walkthrough: create and populate the document

Open BIL481-Project in Gemini and use its **single chat** to request a Google Drive folder named BIL481-2026-Project and one Google Docs document named **Concept Notes** inside it. If the folder already exists, reuse it; create the document **once**. Initialize its seven-section structure, then develop your project proposal using the instructions after the walkthrough. Reuse this document for all Assignment 1 updates.



Screenshot 4. In the notebook's chat, request creation of Concept Notes in the Drive project folder. Open the returned folder and document links to check them.

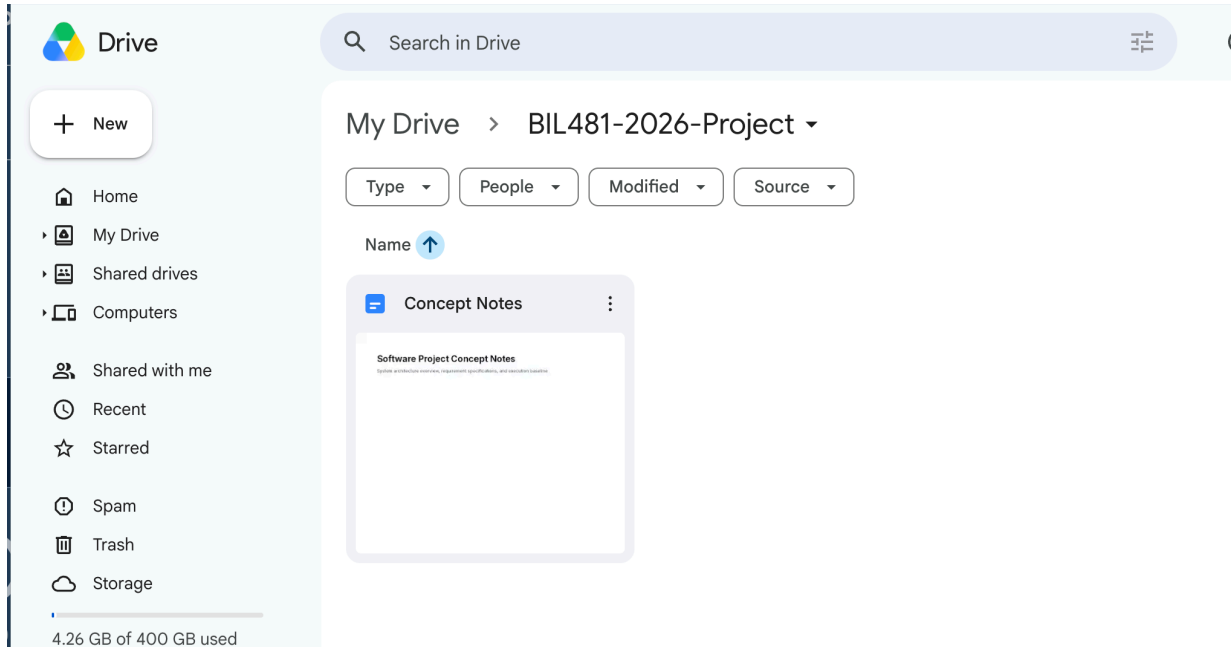


Screenshot 5. In the same chat, ask Gemini to write the required sections into the existing Concept Notes document. Keep later edits in this chat too.

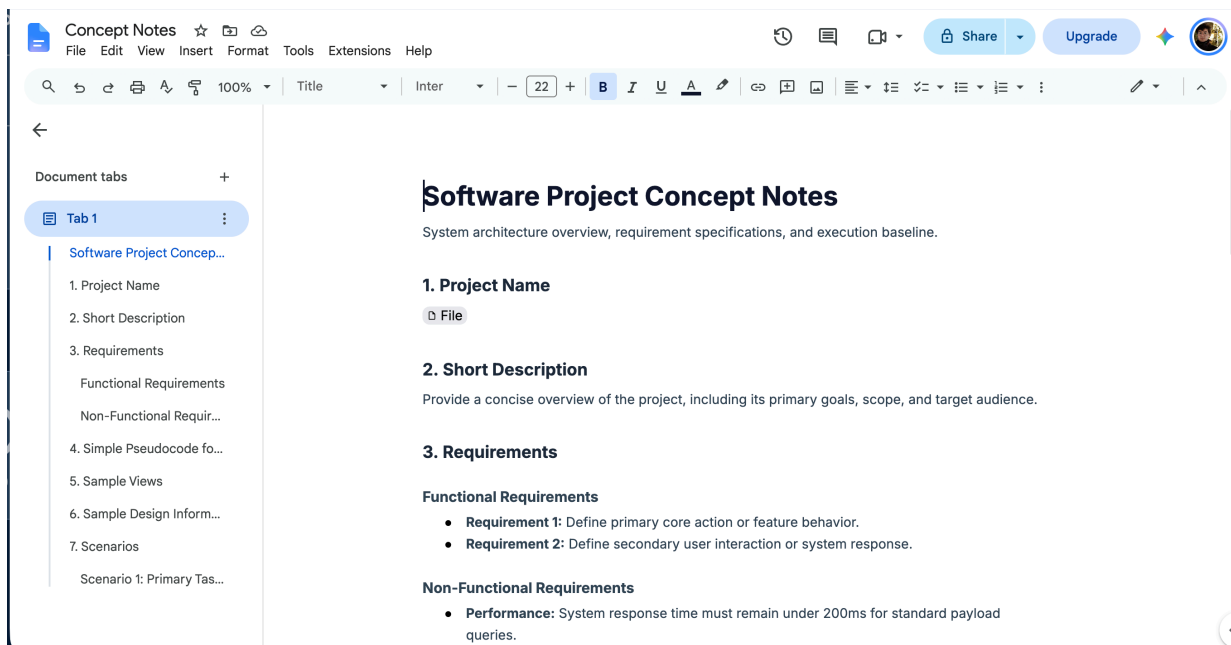
If a required feature is unavailable, contact the TA through Piazza or email and describe the blocked step. Verify actual Drive files and saved content; a success message in Gemini alone does not confirm a saved change.

4 Setup walkthrough: check the document in Drive

Open Google Drive using the same account. Check the folder name, document name, and saved content. The screenshots show an example template; your submission must contain your group's developed project concept. Example performance targets and other placeholder text are not assignment requirements.



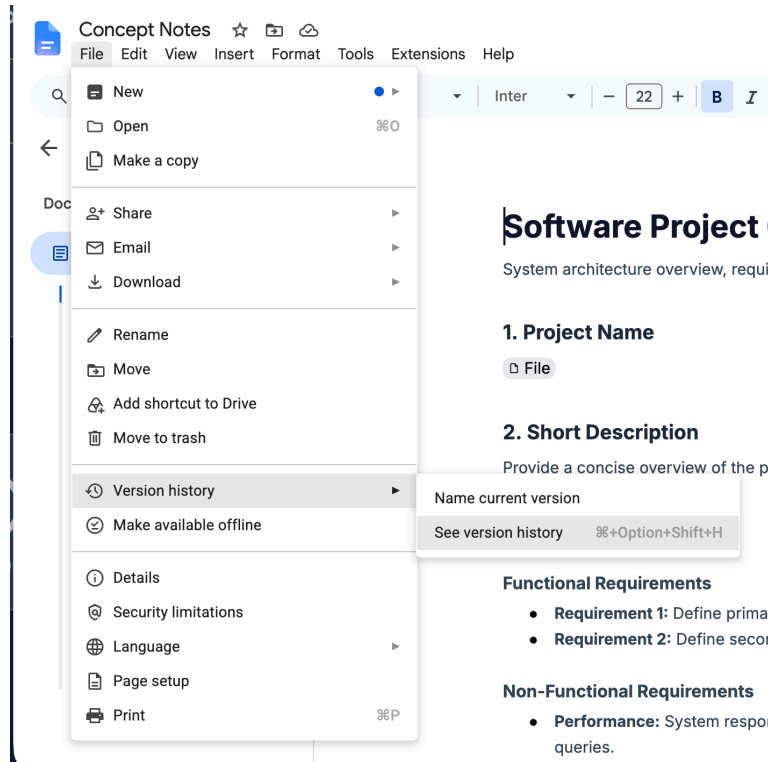
Screenshot 6. Open BIL481-2026-Project in Drive and confirm that it contains the Google Docs document named **Concept Notes**.



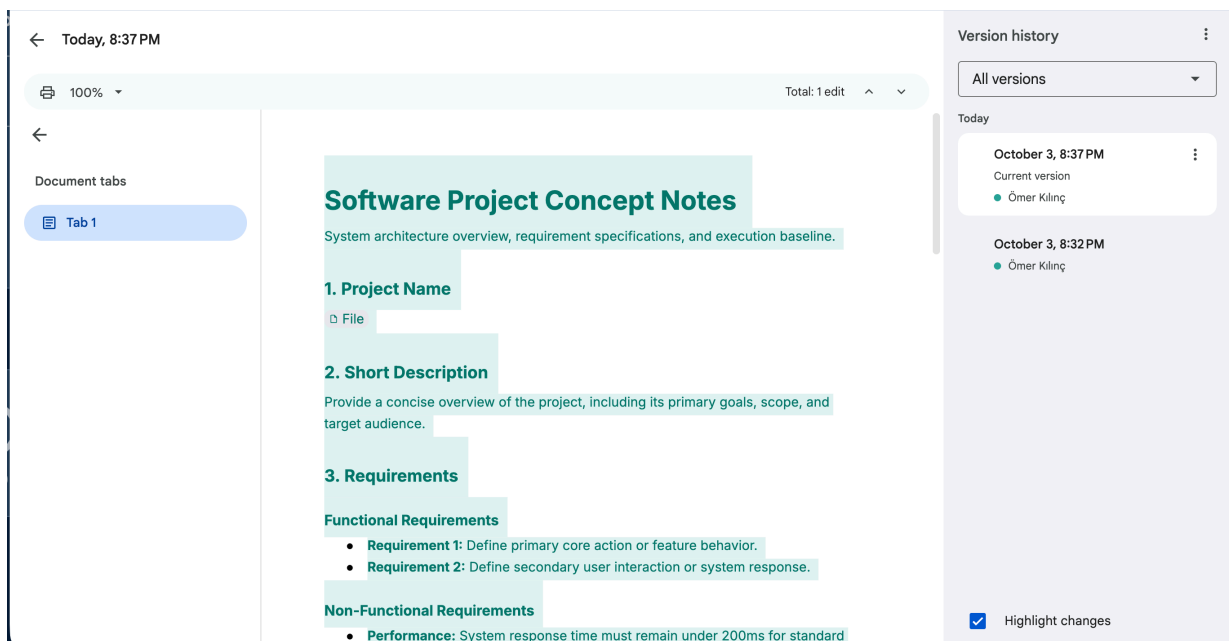
Screenshot 7. Open Concept Notes in Google Docs and check that the requested sections and content were saved. Replace template placeholders with project content through prompts in the same Gemini chat.

5 Setup walkthrough: check saved revisions

After Gemini updates Concept Notes, reopen the document and verify the change. You can inspect earlier saved revisions in Google Docs using **File** → **Version history** → **See version history**.



Screenshot 8. Open **File**, then **Version history**, and select **See version history**.



Screenshot 9. Select a saved revision in the right panel to review its content and highlighted changes. Continue updating the existing Concept Notes document through the original chat in BIL481-Project.

6 Write the initial project prompt

Continue the single Assignment 1 chat in the Gemini notebook named BIL481-Project. Use the **existing Concept Notes document** in your Drive project folder, BIL481-2026-Project.

In the **single chat inside your notebook**, write one prompt describing your group's project idea and asking Gemini to brainstorm its structure, core concepts, and implementation approach. Include the problem, intended users, main goals, and any known constraints.

End the prompt with an explicit command asking Gemini to write the initial proposal into the **existing Concept Notes document**, using all seven sections below. Open it in Drive to verify that the content was saved. **Do not create additional copies of Concept Notes.**

7 Required Concept Notes sections

1. **Project Name.** Give the project a meaningful name. Include your group name and all members' names at the beginning of Concept Notes.
2. **Short Description.** Briefly describe the problem, intended users, purpose, and main idea of the proposed solution.
3. **Requirements.** Include approximately **10 numbered requirements in total**, covering system behavior and relevant quality expectations. Make each requirement clear enough to discuss and check.
4. **Simple Pseudocode for Implementation Parts.** Show the basic logic of the main parts or workflows, including inputs, main steps, and expected outputs.
5. **Sample Views.** Include a few screens or views showing how users interact with the system. Sketches, wireframes, or clearly structured screen layouts are sufficient.
6. **Sample Design Information.** Give preliminary design details, such as main components, data entities, interactions, or technology choices. Explain how they support the concept.
7. **Scenarios.** Include a few concrete usage examples. Identify the user, goal, main steps, and expected outcome in each scenario, keeping them consistent with the requirements and views.

8 Critique, compare, and refine

Continue the **same chat inside BIL481-Project** and make decisions as a group. Keep all Assignment 1 prompts and responses in this one chat.

Critique the proposal. Ask Gemini to identify weaknesses, unclear or missing requirements, design inconsistencies, and feasibility concerns. Discuss the critique and ask follow-up questions about the issues that matter to your project.

Compare similar applications. Ask Gemini to compare your concept with similar applications or approaches. Examine shared features, differences, and opportunities to improve the idea. Check any examples or claims you rely on, and keep the references in the conversation or document.

Refine the concept. Record in the chat which suggestions your group accepts, changes, or rejects, and explain why. **Use command prompts in this same chat to update the existing Concept Notes document in Drive.** For each accepted change, specify the document, project folder, section, and exact revision to apply. Retain all seven required sections, then open the document in Drive to verify that each update was saved.

Example command prompt: "Update the Requirements section of Concept Notes in our Drive project folder with the agreed changes, and save the document."

9 GitHub: Initialize the project repository

Create one public fork per group of the template repository below using GitHub's **Fork** action. One member owns the fork and adds the other group members as collaborators so they can contribute. Clone your fork locally.

<https://github.com/Cpppriest/BIL481-2026-Project>

As part of Assignment 1, complete your fork's `README.md` with your group name and each member's full name, GitHub username, email, and school number. Replace the example entries. Edit the `README` locally and make this the first new commit on your fork; use the exact message Group information and push it. The template's inherited commits do not count as new group commits.

Full Name	GitHub Username	Email	School Number
Ömer Kılınç (<i>example</i>)	@github-username	omerkilinc@etu.edu.tr	261111011
Member 2	@github-username	email@etu.edu.tr	School number
Member 3	@github-username	email@etu.edu.tr	School number
Member 4	@github-username	email@etu.edu.tr	School number
Member 5	@github-username	email@etu.edu.tr	School number

Template `README`: replace the example entries with your group's details.

10 Final checks

Keep the **Concept Notes** document only in your project folder on Google Drive. Do not upload it elsewhere. Ensure that the course staff have viewing access. Open the document in Drive and verify that it is complete and readable.

Keep the **complete Assignment 1 chat history** in the single chat inside BIL481-Project: all setup messages, the exact initial project prompt and response, and every follow-up prompt and response used for critique, comparison, and refinement.

Before submitting, check that the requirements, pseudocode, views, design information, and scenarios describe the same project. The chat should show how the concept evolved, and the final document should reflect your group's decisions. Every member should be able to explain the concept and its main design choices.

11 Submission

Complete the following by the deadline on the first page. Identify your group and all members clearly.

1. **Submit the GitHub fork link.** Provide the URL of your group's public fork of [the project template repository](#). Its README.md must contain your group's details, and its first new group commit must have the exact message Group information. Confirm that this commit has been pushed to the fork.
2. **Share the Gemini chat link.** Share your group's complete Assignment 1 chat in the notebook BIL481-Project. Include all setup messages, the initial project prompt, and all subsequent prompts and responses. Below the final response, select **Share** → **Share conversation** and copy the public link. Generate the share link **after completing all Assignment 1 prompts and document updates**. Open the shared link and verify that the conversation is complete through the final response. If you continue the conversation, generate an updated share link before submitting.
3. **Share the Google Drive folder link.** Share your BIL481-2026-Project folder containing the final **Concept Notes** Google Docs document. Ensure that all group members and course staff can open the folder and view the document. Share viewing access with omerkilinc@etu.edu.tr and the instructor's course account.

Where to submit all three links (GitHub fork, Gemini chat, and Google Drive folder):
To be announced. Further submission instructions will be announced later.

12 Restrictions

1. **Concept Notes must not exceed 10 pages in total.** This includes all sections, figures, and any title or reference pages. Use a paginated layout in Google Docs with **A4 pages**, a body font of **at least 11 pt**, and margins of **at least 2 cm** on all sides, so the page count can be checked consistently.
2. **The notebook workflow is required for every group.** Whether you use Google AI Plus or an existing Pro or Ultra subscription, use BIL481-Project and **one chat** for all Assignment 1 prompts and document edits, as described in Setup. **Do not use Spark for Assignment 1.**

13 Questions and office hours

Use the course's **Piazza Q&A section** for assignment questions. You may also contact the TA directly by email or in person during the office hours below.

Teaching assistant	Ömer Kılınç
Email	omerkilinc@etu.edu.tr
Office hours	Thursday, 16:30–18:20; Friday, 16:30–18:20
Location	TM-213